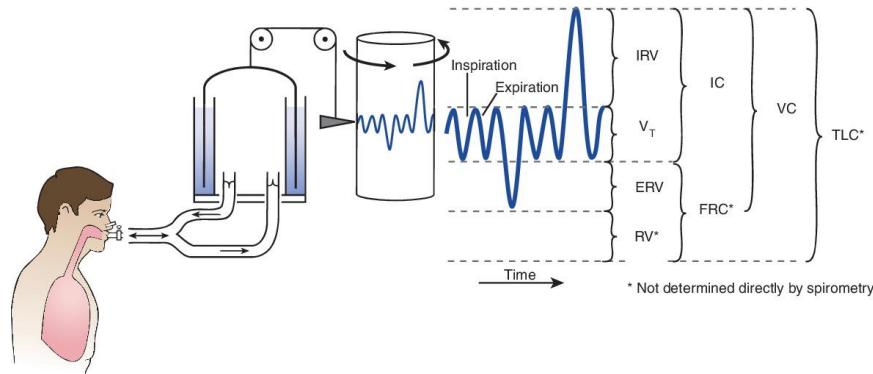


Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture No: 31

Spirometry



* Not determined directly by spirometry

Rhoades and Bell, *Medical Physiology: Principles for Clinical Medicine*, Fourth Edition, ©Lippincott, Williams & Wilkins (2013), pp. 335

Dr. Esther Johnson
St Georges University ©
ejohnso3@sgu.edu

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Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0221 Draw a normal spirogram, labeling the four lung volumes and four capacities. List the volumes that comprise each of the four capacities. Identify which volume and capacities cannot be measured by spirometry.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0222 Indicate the value of using the Helium Dilution test and Plethysmography.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0223 Explain the factors that determine total lung capacity, functional residual capacity, and residual volume.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0366 Describe the mechanisms responsible for the changes in the lung volumes and capacities that occur in patients with emphysema and pulmonary fibrosis.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0224 Draw a spirogram resulting from a maximal expiratory effort and label the forced vital capacity (FVC), timed forced expiratory volumes (FEVs), and the maximal expiratory flow rate between 25-75% of FVC (FEF25-75%).

SOM.MK.II.BPM1.3.CPR.2.PHYS.0367 Describe how the mechanical forces that contribute to the pulmonary function indices alter with obstructive or restrictive disease.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0225 Identify the effort-dependent and -independent regions on a normal maximal effort flow-volume curve.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0368 Describe how dynamic compression of airways influences the shape of the effort-independent region of the maximal effort flow-volume curve .

SOM.MK.II.BPM1.3.CPR.2.PHYS.0369 Describe how and why the shape of the flow-volume curve is shifted in chronic obstructive and restrictive lung diseases.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0226 Differentiate between the two broad categories of restrictive and obstructive lung disease, including the spirometric abnormalities associated with each category.

Assigned Reading

- Rhoades and Bell, *Medical Physiology: Principles for Clinical Medicine*, Fifth Edition, © Lippincott, Williams & Wilkins (2018), Chapter 18
<https://meded.lwwhealthlibrary.com.periodicals.sgu.edu/content.aspx?sectionid=165535373&bookid=2188#165535437>

Negative Intrapleural Pressure (Concept Review)

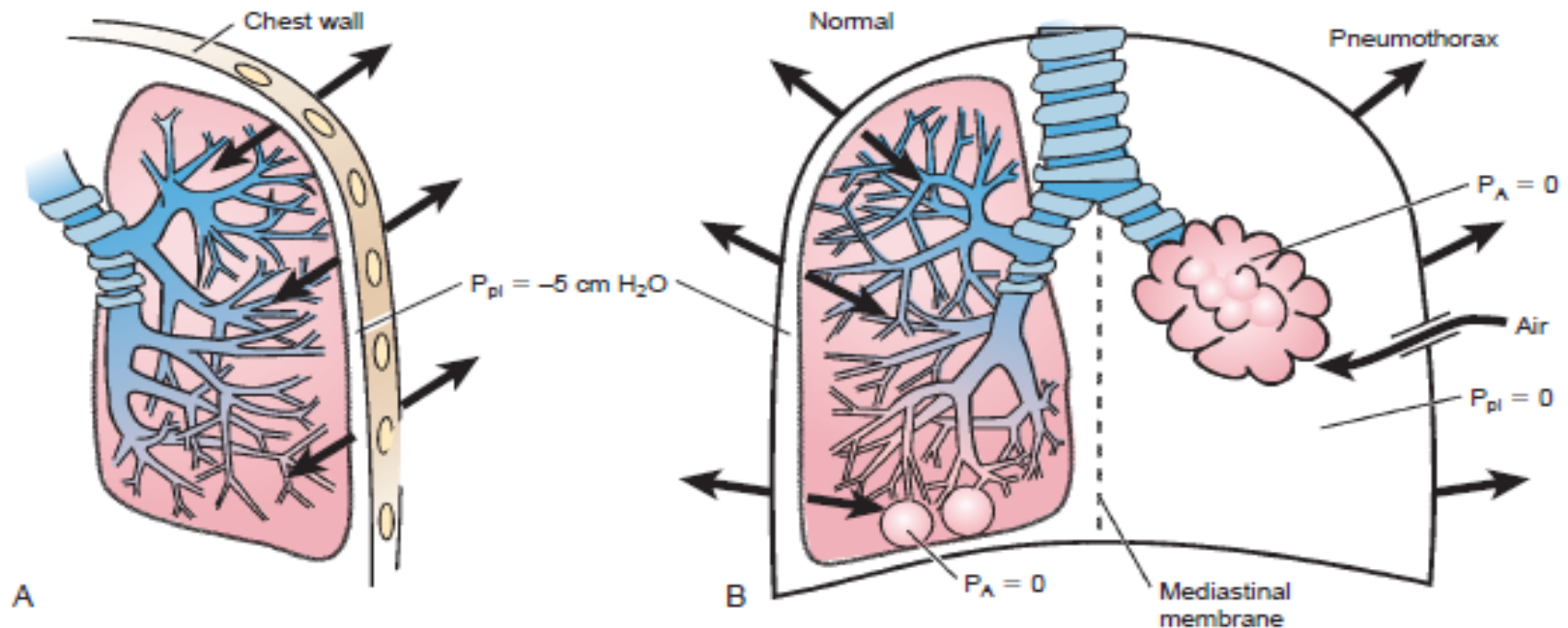
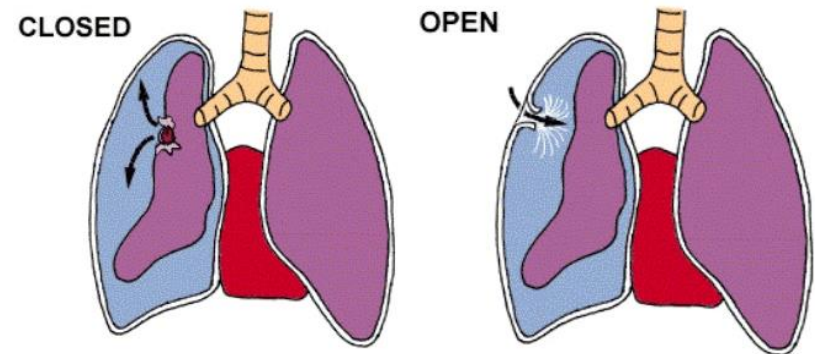


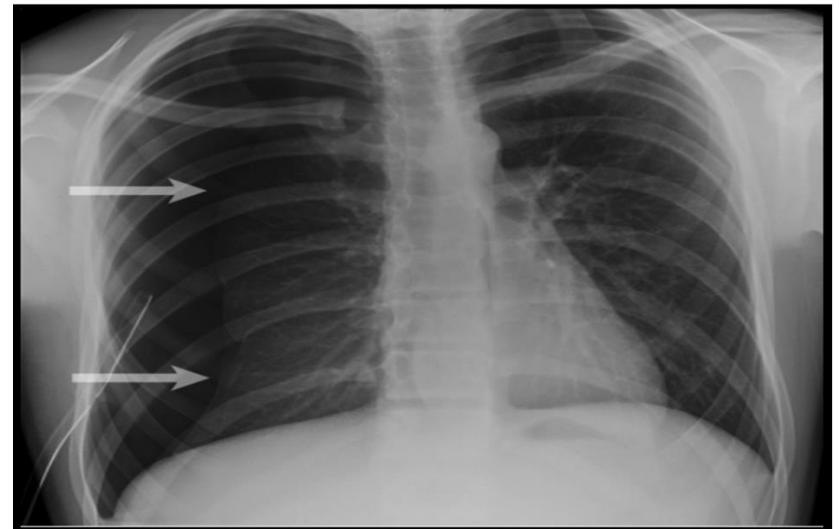
Figure 18.5 A negative pleural pressure results from the elastic recoil of the lungs and chest wall pulling in opposite directions. **(A)** The stretched lung (at the end of a normal expiration) tends to recoil inwardly, and the chest wall tends to recoil outwardly, but in equal and opposite directions. Consequently, pleural pressure (P_{pl}) becomes negative (i.e., less than atmospheric pressure). **(B)** Rupture or puncture of the lung or chest wall results in a pneumothorax, during which the transpulmonary pressure becomes zero and elastic recoil causes the lung to collapse. The mediastinal membrane prevents the other lung from collapsing. P_A , alveolar pressure.

Clinical Consideration: Pneumothorax

- The presence of air or gas in the pleural cavity
- Tension Pneumothorax
 - cause a shift of the mediastinum and compromise hemodynamic stability
- Causes
 - Spontaneous
 - Traumatic



http://www.fprmed.com/Medical/Pulm/pneumo_Open_vs_Closed.gif



<http://studymedicalphotos.blogspot.com/2015/05/a-25-year-old-man-presents-with-3-day.html>

Negative Intrapleural Pressure

Recoil – ability to return to resting volume once stretched

- **Lung Recoil** – inward force produced by the elastic properties of the alveoli; acts to collapse the lung.
- **Chest wall recoil (PCWR)** – outward force produced by the chest wall

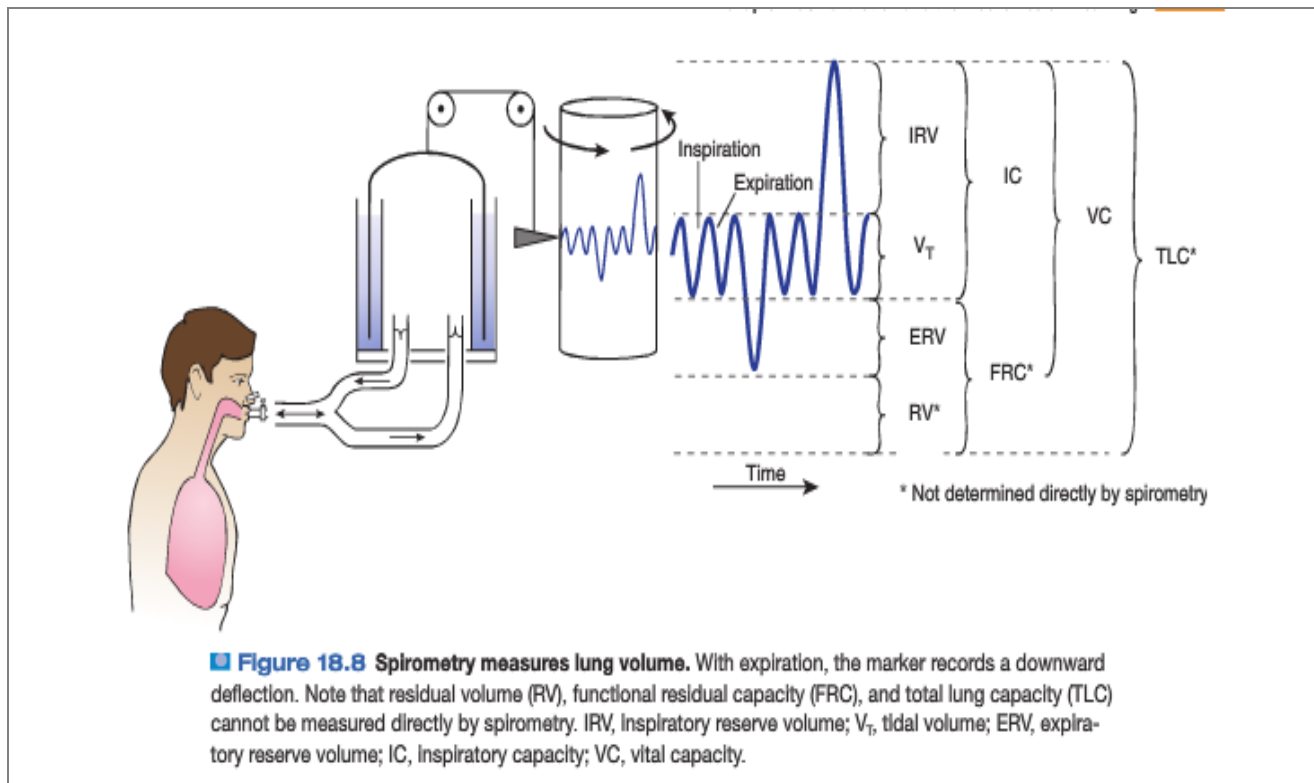
Intrapleural Pressure (P_{IP})

- The pressure within the pleural cavity.
- The opposing recoil forces of the chest wall and the lung creates a negative (subatmospheric) P_{IP} .

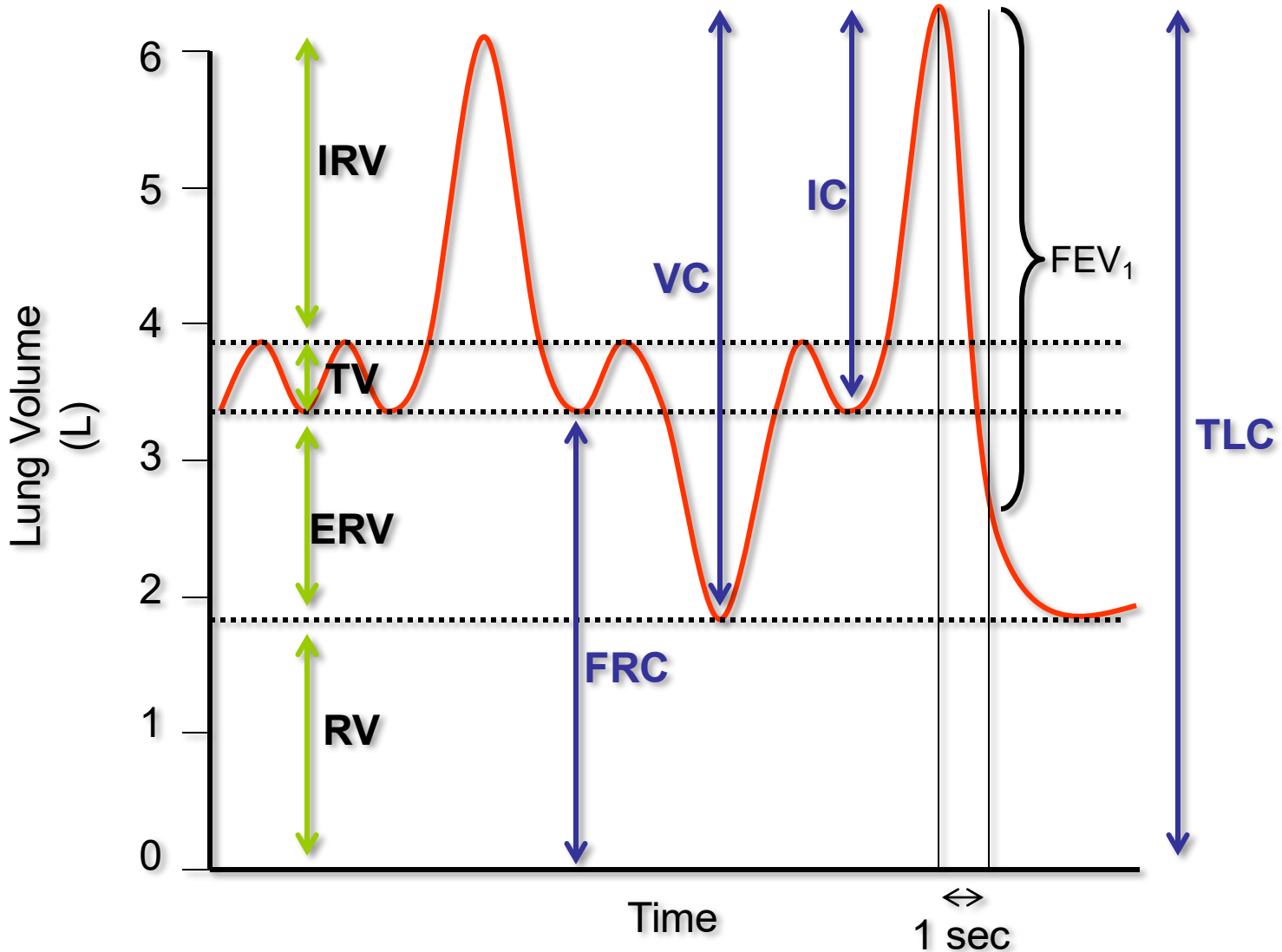
pulmonary function test (PFT)

Spirometry – The Spirometer

One the most common of the pulmonary function tests used to assess lung function



Spirometry – The Spirogram



Spirometry – Terminology

- **Tidal Volume (TV):** the volume of air inhaled or exhaled with each normal breath
- **Inspiratory Reserve Volume (IRV):** the volume of air that can be inhaled at the end of a normal tidal inspiration
- **Expiratory Reserve Volume (ERV):** the volume of air within the lungs that can be exhaled after the end of a tidal exhalation
- **Residual Volume (RV):** the air remaining in the lungs after a maximal expiration (this volume of gas cannot be expelled, and cannot be measured by spirometry)

Spirometry – Terminology

- **Functional Residual Capacity (FRC):** the total volume of air remaining in the lungs at the end of a tidal exhalation
- **Total Lung Capacity (TLC):** the volume of air in the lungs at the end of a maximal inspiration
- **Vital Capacity (VC):** the volume of air exhaled from maximal inspiration to maximal exhalation (maximum expiration). When done with force this volume is termed the **forced vital capacity (FVC)**
- **Forced Expiratory Volume in 1 second (FEV1):** the volume of air exhaled in the first second of a FVC test

Lung Volumes and Capacities

Abbr.	Volume/Cap.	Vol. Range (L)
IRV	Inspiratory Reserve Volume	1.9 – 2.5
TV	Tidal Volume	0.4 – 0.5
ERV	Expiratory Reserve Volume	1.1 – 1.5
RV	Residual Volume	1.5 – 1.9
TLC	Total Lung Capacity ($TLC = IRV + VT + ERV + RV$)	4.9 – 6.4
IC	Inspiratory Capacity ($IC = VT + IRV$)	2.3 – 3.0
FRC	Functional Residual Capacity ($FRC = ERV + RV$)	2.6 – 3.4
VC	Vital Capacity ($VC = IRV + ERV + VT$)	3.4 – 4.5

Lung volumes/ capacities vary with - age, gender, race & height

Take home Test:

Which of the following terms best describe the volume of air remaining in the lung at the end of FVC?

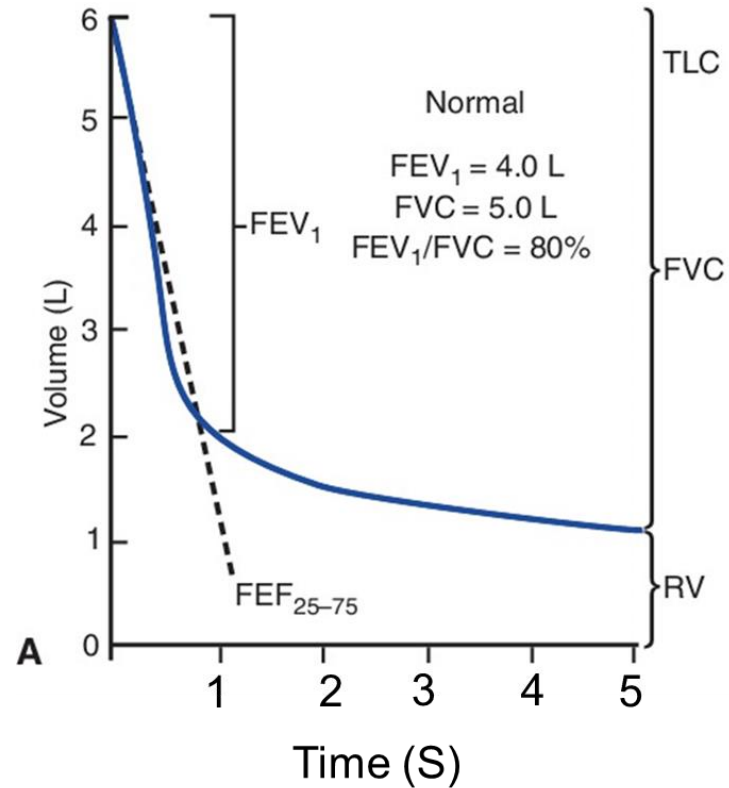
- ✓ A. RV
- B. IC
- C. ERV
- D. TV
- E. FRC

Explanation: Vital Capacity (VC) is the volume of air exhaled from maximal inspiration to maximal exhalation; maximum expiration. When done with force this volume is termed the **forced vital capacity (FVC)**. Therefore, the remaining air in the lung can't be expired anymore and is known as the residual volume (RV). This RV can't be directly measured by the spirometry.

Spirometry – FVC & FEV1

Spirometry- Use of FEF₂₅₋₇₅

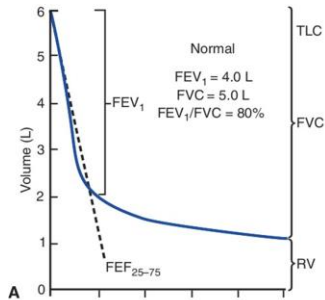
- Forced Expiratory Flow (FEF)
- FEF₂₅₋₇₅ Represents the expiratory flow rate over the middle half of the FVC (between 25% and 75%).
- Small airway obstruction may be present even when the FEV₁/FVC% is above the lower limit of normal.
- FEF₂₅₋₇₅ has the greatest sensitivity for the detection of early (distal) airflow obstruction.



Normally FEV₁ should be ~80% of total volume expired (FVC)

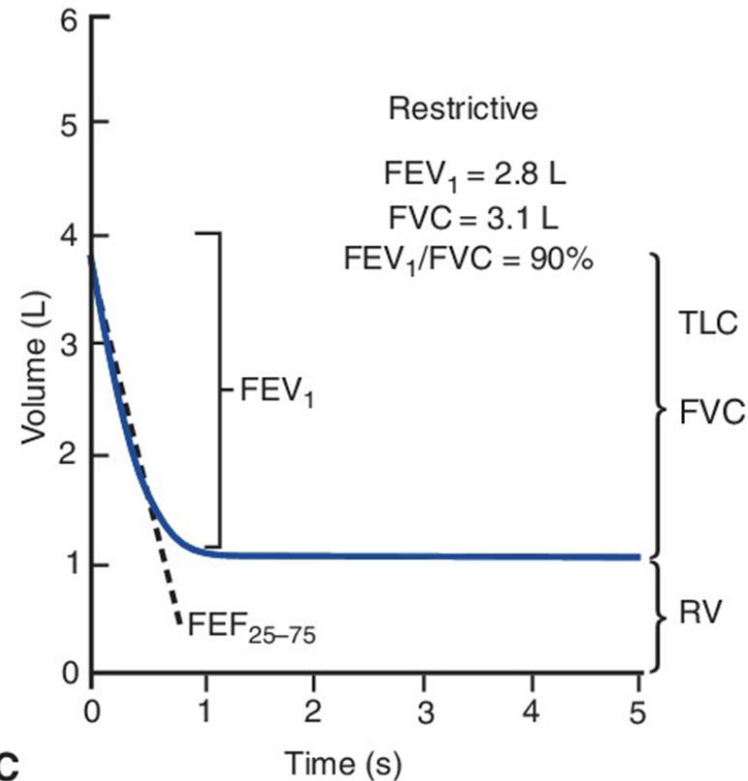
$$\text{FEV}_1/\text{FVC ratio} = 4 \text{ L}/5 \text{ L} * 100 = \sim 80\%$$

Spirometry – Restrictive Lung Disease



For Normal person,
 $FEV_1/FVC =$
 $4/5 \times 100 = \sim 80\%$

Patient with restrictive lung disease



C

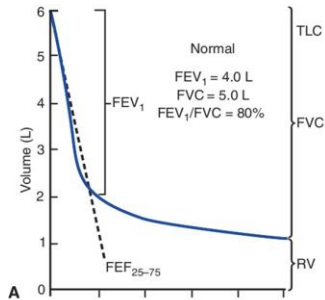
- Both FVC & FEV₁ is reduced

- The FEV₁/FVC ratio remains normal or may be increased.

For patient,
 $FEV_1/FVC =$
 $2.8/3.1 \times 100 = \sim 80\%$

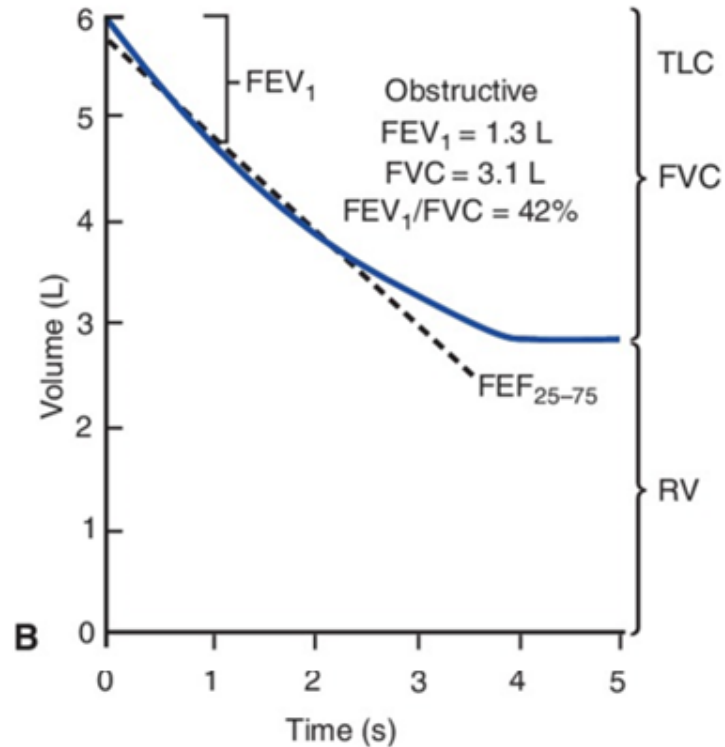
Spirometry – Obstructive Lung Disease

Patient with obstructive lung disease



For Normal
person, FEV_1/FVC
 $= 4/5 \times 100 =$
 $\sim 80\%$

- FEV₁ is reduced significantly
- FVC is reduced
- FEV₁/FVC ratio <0.8 or <80%



For patient,
 $FEV_1/FVC =$
 $1.3/3.1 \times 100 = \sim 42\%$
 $<80\%$

Changes in Emphysema & Pulmonary fibrosis

Parameter	Emphysema (Obstructive)	Pulmonary Fibrosis (Restrictive)
Total Lung Capacity (TLC)	Increased: Due to destruction of alveolar walls, causing lung hyperinflation and air trapping.	Decreased: Lung tissue becomes stiff and less compliant, limiting lung expansion.
Functional Residual Capacity (FRC)	Increased: Air trapping leads to a higher volume of air remaining in lungs after normal exhalation.	Decreased: Loss of lung compliance reduces resting lung volume.
Residual Volume (RV)	Increased: Damaged alveoli reduce elastic recoil, making it harder to fully exhale.	Decreased or Normal: Less air remains after exhalation due to overall smaller lung size.
Vital Capacity (VC)	Decreased or Normal: Loss of elastic recoil hinders complete exhalation.	Decreased: Lung stiffness reduces the maximum volume of air exhaled after a deep breath.
Forced Expiratory Volume (FEV1)	Significantly Decreased: Airway collapse and decreased elastic recoil slow expiration.	Decreased or Normal: Reduced lung volume limits airflow but not as drastically as in obstructive diseases.
FEV1/FVC Ratio	Decreased: Due to a greater reduction in FEV1 compared to Forced Vital Capacity (FVC).	Normal or Increased: Both FEV1 and FVC are reduced, often proportionately, maintaining or slightly increasing the ratio.

Spirometry – FVC & FEV1

- From total lung capacity there is forceful exhalation into the spirometer.
- Measurements of FVC, FEV1 and FEV1/FVC ratio are obtained and used to detect obstructive and restrictive disorders.

Obstructive disorder

- Expiratory flow rate is significantly decreased resulting in decreased FEV1 and FVC
- The FEV1/FVC ratio is low.

Restrictive disorder

- Lung inflation is decreased resulting in decreased FEV1 and FVC
- The FEV1/FVC ratio is normal or increased.

Obstructive and Restrictive Lung Diseases

Obstructive Disease	Restrictive Disease
Expiratory flow rate decreased	Lung inflation is decreased
Airway occlusion, air trapping	<u>Intrinsic</u> - lung tissue <u>Extrinsic</u> – chest wall ,pleura or neuromuscular
Examples: 1. Chronic Bronchitis 2. Emphysema 3. Asthma	Examples : 1. Pulmonary Fibrosis 2. IRDS 3. Scoliosis

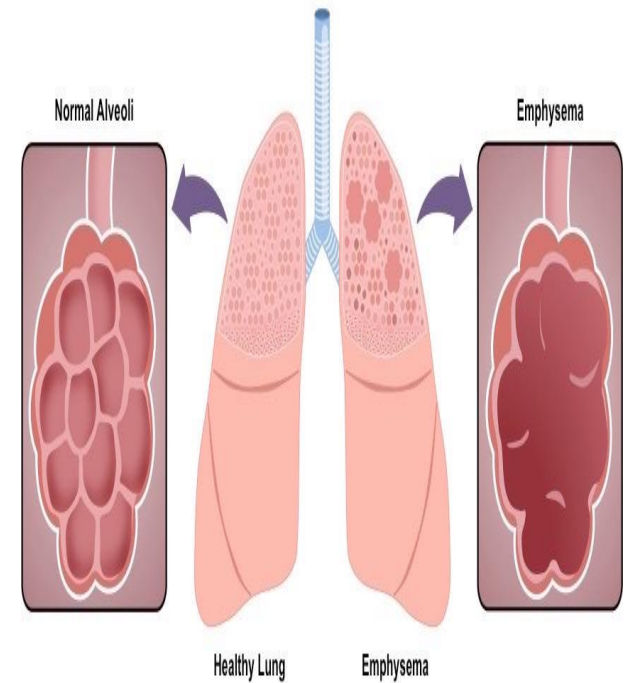
Spirometry : Obstructive and Restrictive Disease

Volume	Obstructive Disease	Restrictive Disease
FVC	↓	↓ ↓
FEV ₁	↓ ↓	↓
FEV ₁ /FVC ratio	<0.8	≥0.8
TLC	↑	↓
RV	↑	↓
FRC	↑	↓

Characteristics of obstructive and restrictive lung disease

Clinical Consideration – Emphysema (Review)

- Disintegration of the lung's elastic framework and destruction of alveolar walls
- Enlargement of air spaces distal to the terminal bronchioles
- Signs of air trapping – prolonged expiratory phase, barrel chest, flat diaphragm
- Commonly caused by **cigarette smoking**
- Genetic risk factor - **alpha-1-antitrypsin (AAT) deficiency**



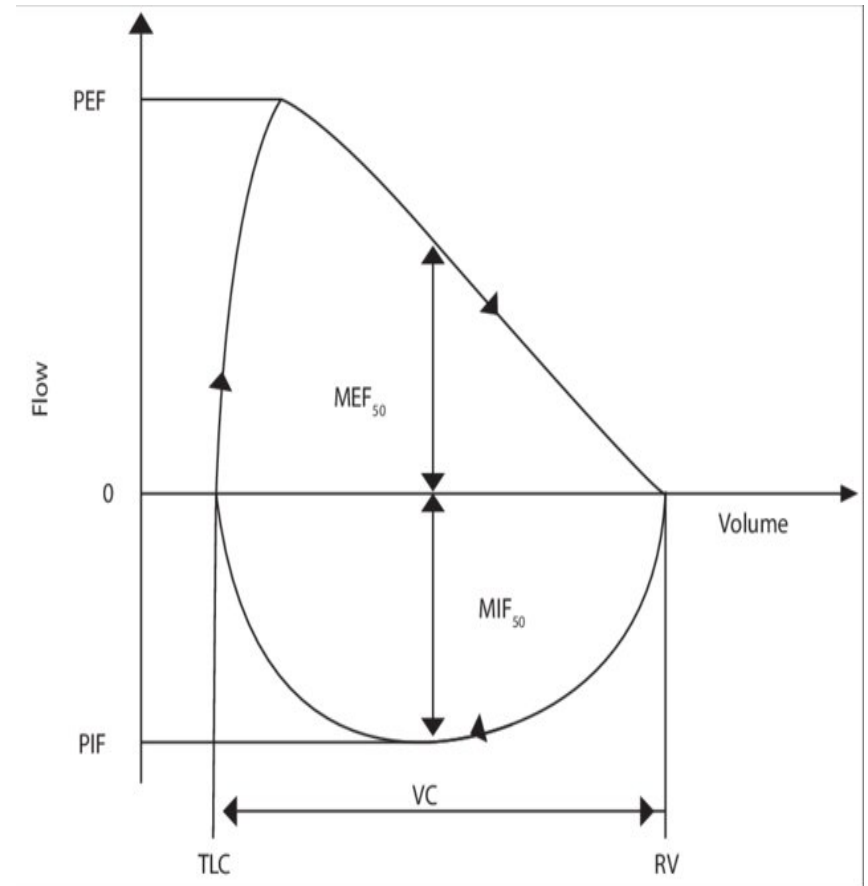
http://ib.bioninja.com.au/_Media/emphysema-2_med.jpeg

Pulmonary Function Test

Spirometry-Flow-Volume Loop

Normal flow volume loop.

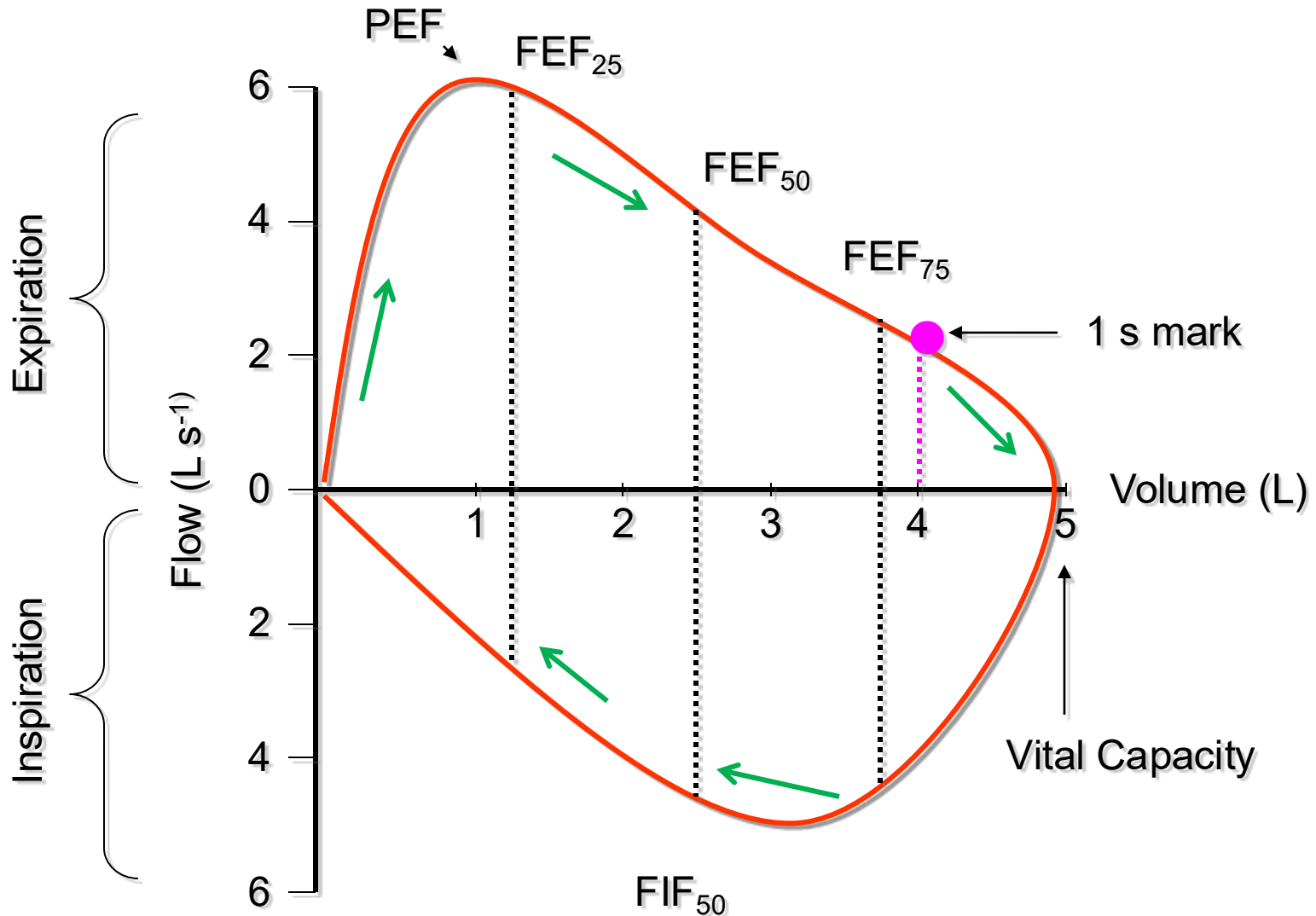
PEF = peak expiratory flow; **PIF** = peak inspiratory flow;
MEF 50 = maximal expiratory flow after 50% of expired forced vital capacity (FVC);
MIF 50 = maximal inspiratory flow after 50% of expired FVC;
TLC = total lung capacity;
VC = vital capacity;
RV = residual volume



Pediatric spirometry guideline of the South African Thoracic Society: Part 1
https://www.researchgate.net/publication/259153833_Paediatric_spirometry_guideline_of_the_South_African_Thoracic_Society_Part_1

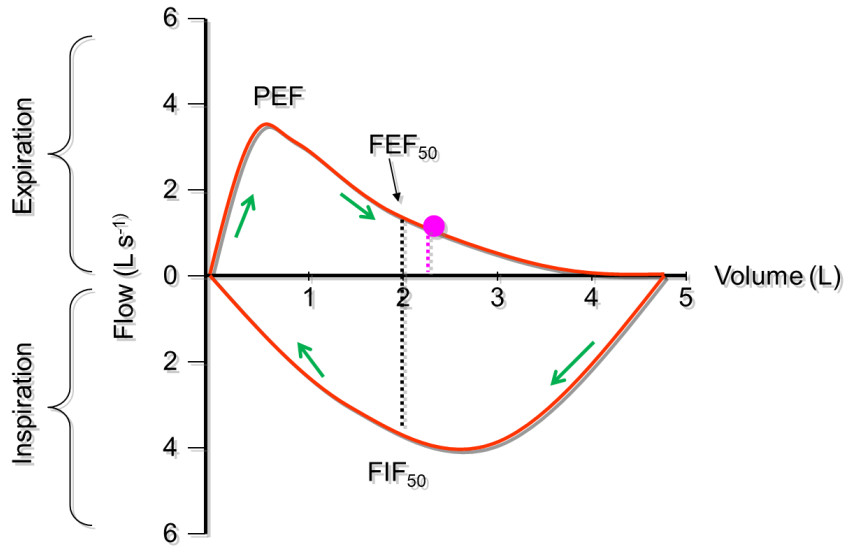
Pulmonary Function Test

Spirometry-Normal Flow-Volume Loop



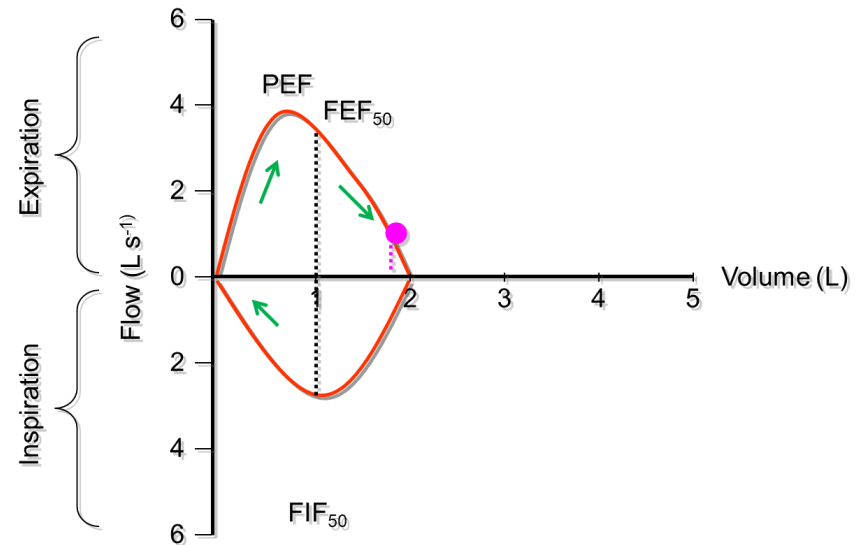
Spirometry- Flow-Volume Loop

Obstructive Vs. Restrictive Lung Disease



Obstructive Disease
Scooped out appearance
common in emphysema

● 1 sec



Restrictive Lung Disease

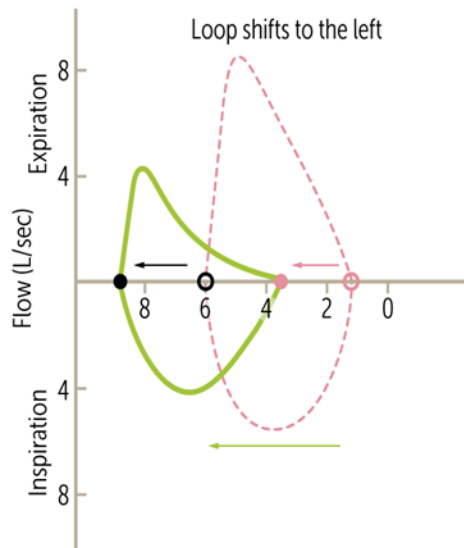
- Steep expiratory limb
- Significant reduction in both FEV₁ & FVC

Spirometry- Flow-Volume Loop

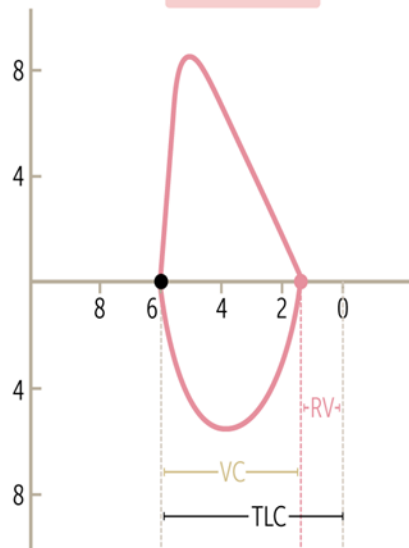
Restrictive and Obstructive Lung Disease (DIY task)

Flow volume status of two patients (1 and 2) of lung disorder is given along with the normal person. Identify and explain the physiology of the two patients compared to the normal healthy person

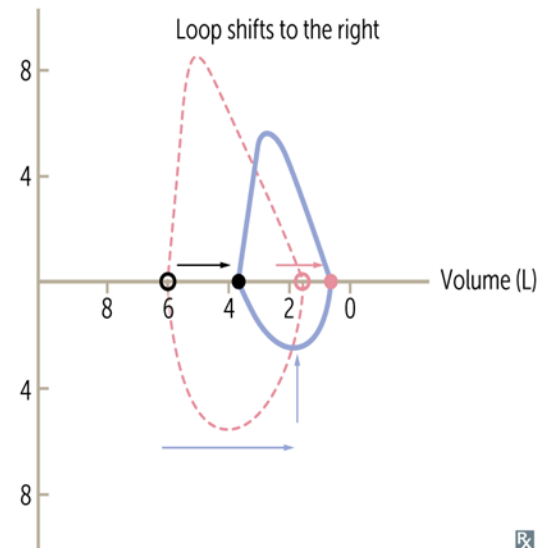
Patient 1



NORMAL



Patient 2



Spirometry- Flow-Volume Loop

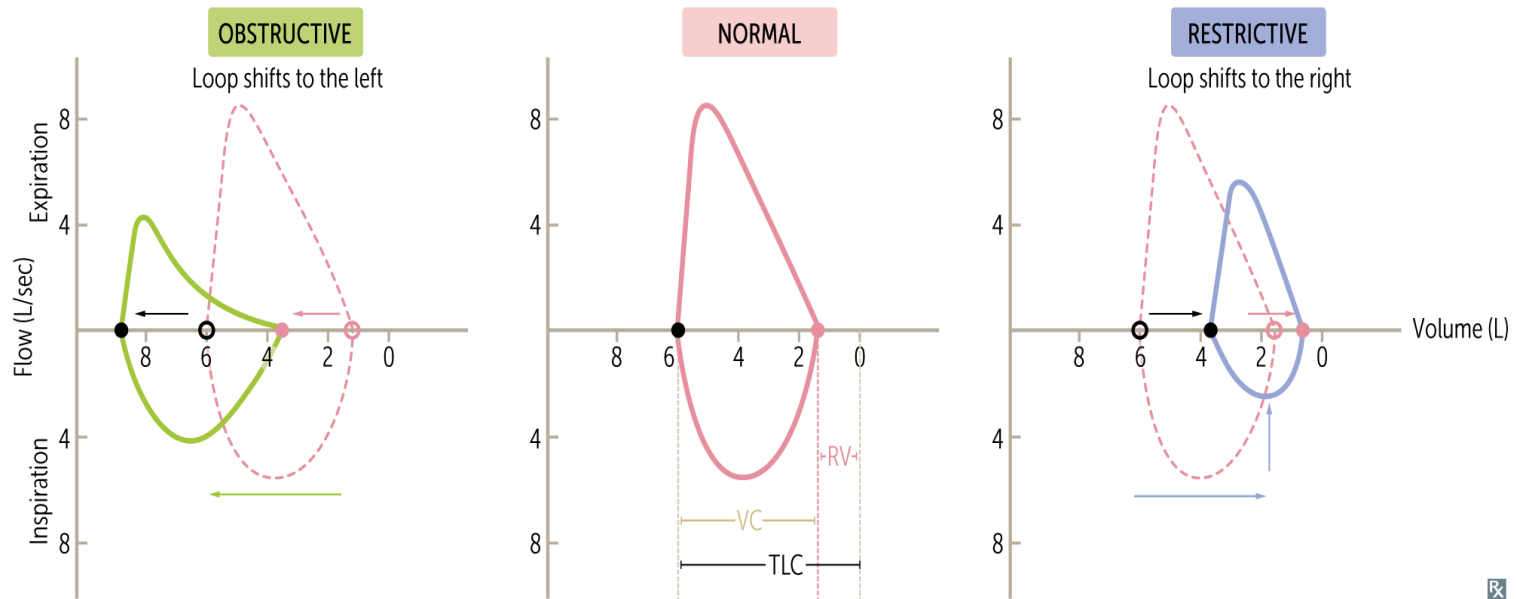
Restrictive and Obstructive Lung Disease

(Explanation of the DIY task)

FA17 p 637.2

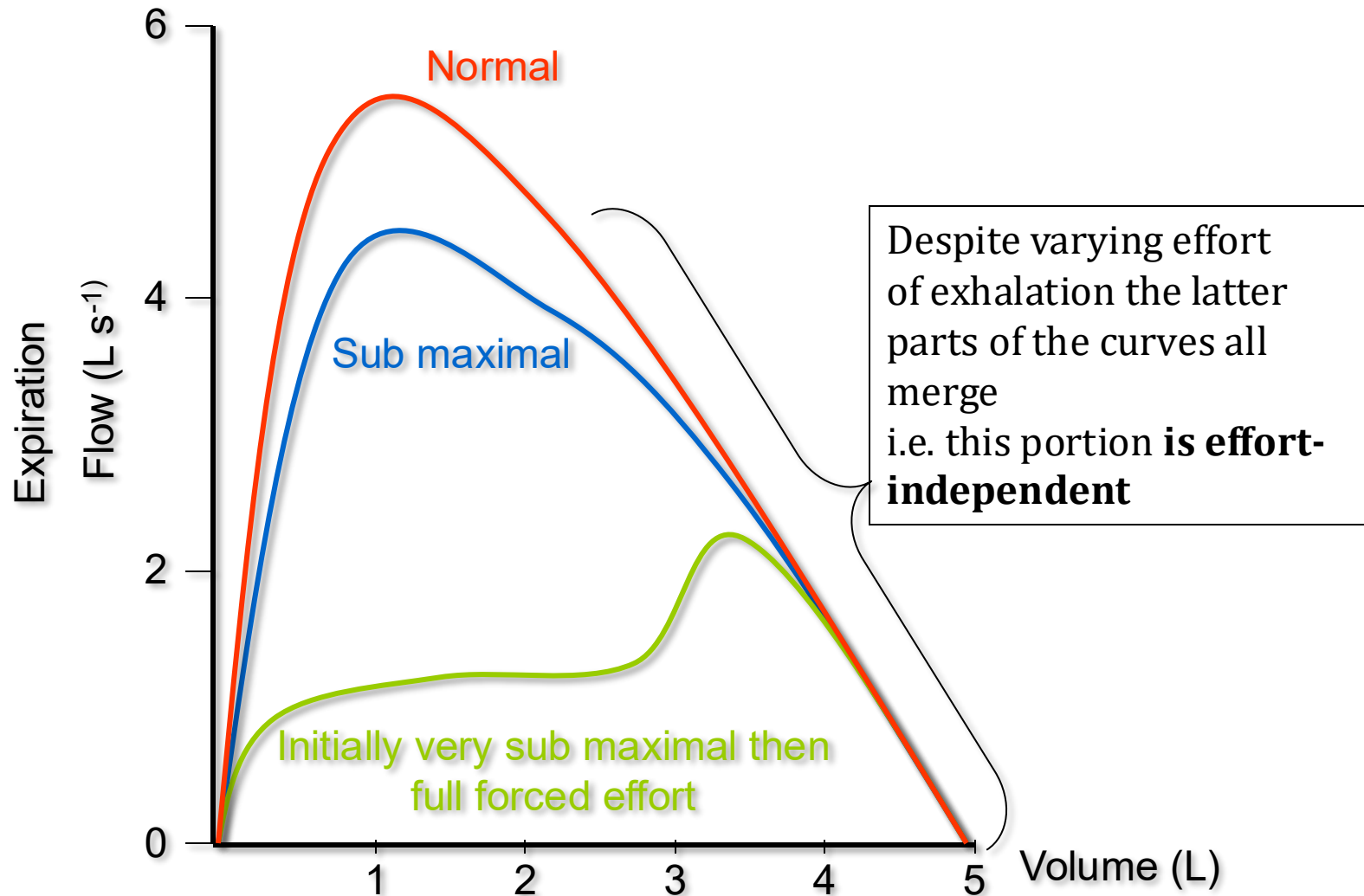
Flow volume loops

Obstructive lung volumes > normal ($\uparrow$ TLC, $\uparrow$ FRC, $\uparrow$ RV); restrictive lung volumes < normal. In obstructive, FEV_1 is more dramatically reduced compared with FVC $\rightarrow$ decreased FEV_1/FVC ratio. In restrictive, FVC is more reduced or close to same compared with $FEV_1 \rightarrow$ increased or normal FEV_1/FVC ratio.



Spirometry- Flow-Volume Loop

Effort Dependence/Independence



Spirometry- Flow-Volume Loop

Effort Dependence/Independence

- The curve is **effort dependent** until PEF is achieved.
- Regardless of effort the, flow after PEF is limited due to compressed airways! (**Effort independence**)
- **Dynamic Compression of Airways** increases resistance to airflow.
- This effect is magnified in lungs with low elastic recoil (high compliance) –airways are floppy, and the lung recoil does little to augment airway pressures above +ve intrapleural pressure during forced expiration.

Clinical Correlation: COPD

- Chronic obstructive pulmonary disease (COPD)
- Combination of chronic bronchitis, emphysema and asthma
- The most common chronic lung disease in the United States
- Commonly caused by cigarette smoking.
- The classical feature is decreased airflow due to chronic obstruction of the small airways
- Results in a decrease in FVC, FEV1 and FEV1/FVC ratio
- In severe cases air is trapped in the lungs during forced expiration

Clinical Correlation: Obstructive Diseases

Clinical findings	COPD	Asthma
Age of onset	Above 35 yrs.	Any age
Onset of dyspnea	Gradual	Sudden
Productive cough	Years	With attack
Childhood wheeze	Uncommon	Common
Family history	Uncommon	Common
Smoking history	Yes	Variable
FEV1 and FVC	Reduced	Reduced during an attack
FEV1:FVC	Reduced	Reduced during an attack
VC	Reduced	Nearly normal
Response to steroids and bronchodilator	Minimal	Strong

Clinical Correlation: Obstructive Diseases

Peak Flow Device

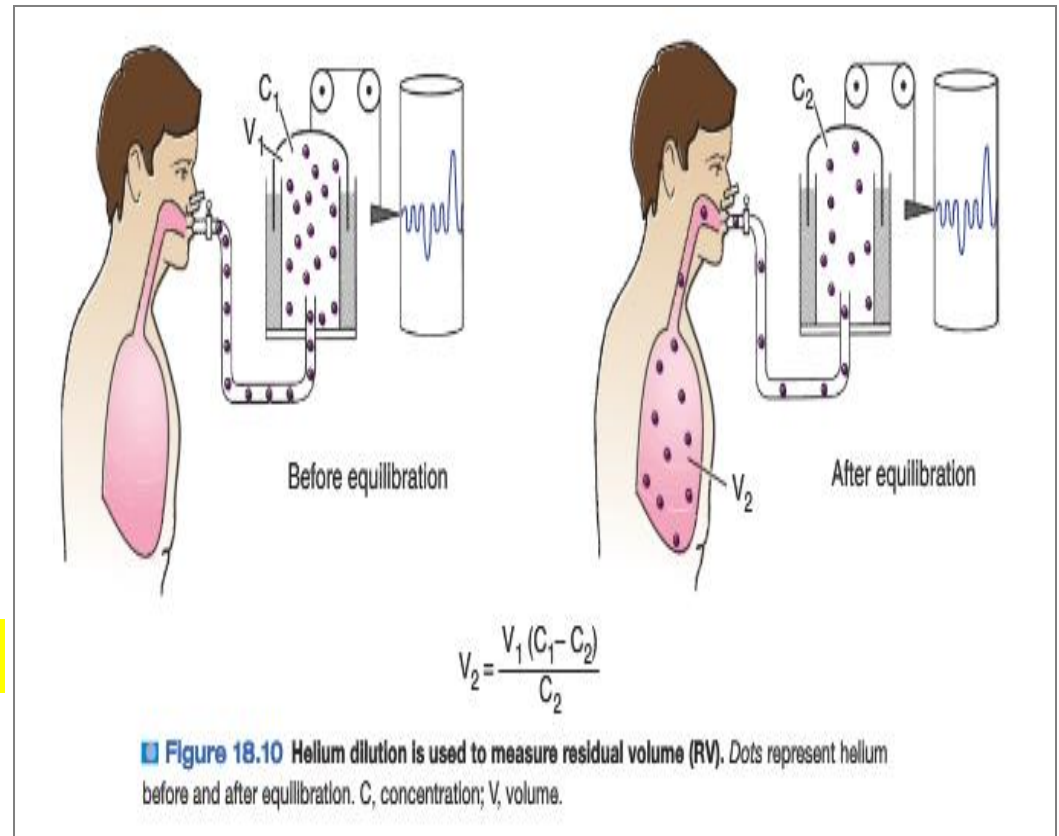
- Used for monitoring of lung disease
- Provides information about exacerbations
- Objective data is collected by the practitioner



http://www.peakflow.com/top_nav/meter/index.html

Other Tests – Helium dilution Technique

- Useful in healthy individuals with no lung obstruction
- Helium does not get beyond obstruction, thus yielding a false FRC
- Body plethysmography can overcome this problem



Rhoades and Bell, *Medical Physiology: Principles for Clinical Medicine*, Fourth Edition, © Lippincott, Williams & Wilkins (2013), pp.338

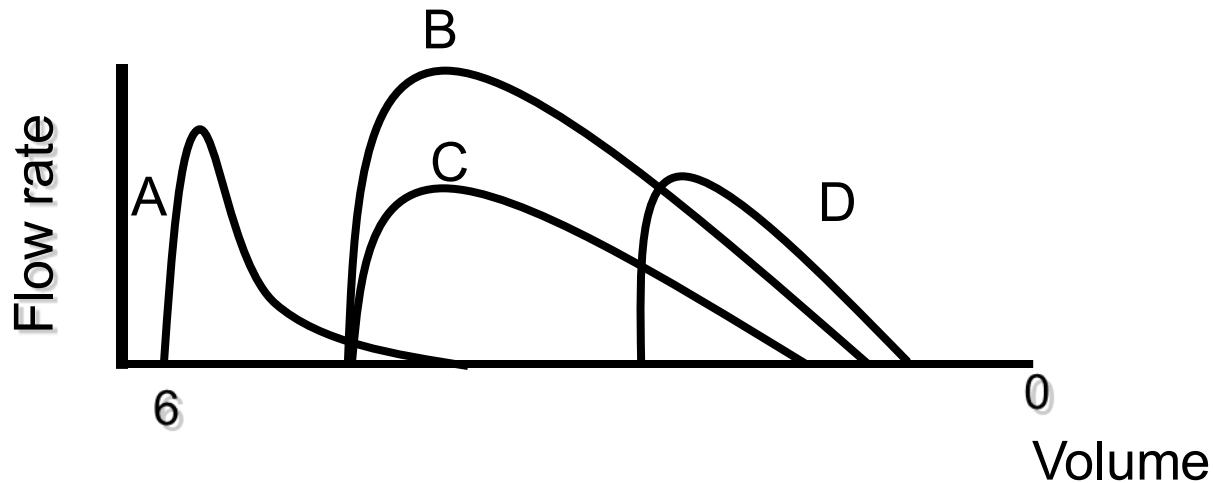
Other Tests-Body Plethysmograph

- Body box
- Airtight box that measures changes in pressure and volume on inspiration and expiration
- Boyle's law is used to estimate lung volume.



Bonus DIY

A 65-year-old man comes to his physician because he has been having difficulty breathing. He has smoked approximately two packs of cigarettes a day for 40 years. His breath sounds are decreased bilaterally, there is hyperresonance on percussion and a recent chest X-ray shows a barrel chest and flat diaphragm. Which of the expiratory flow-volume curves would be expected for this patient?



- A. A
- B. B
- C. C
- D. D

Summary

- The opposing recoil forces of the chest wall and the lung creates a negative (subatmospheric) P_{IP} .
- RV, FRC and TLC cannot be measured with spirometry
- Helium Dilution and Body Plethysmography are used to assess RV, TLC and FRC. Body Plethysmography is better suited for measuring these volumes in persons with obstructive airway diseases
- In restrictive diseases lung inflation is decreased resulting in decreased FEV1 and FVC .The FEV1/FVC ratio is normal or increased.
- In obstructive diseases expiratory flow rate is significantly decreased resulting in decreased FEV1 and FVC .The FEV1/FVC ratio is low.

End of Spirometry

Thank you

For questions and clarifications

please email

ejohnso3sgu.edu